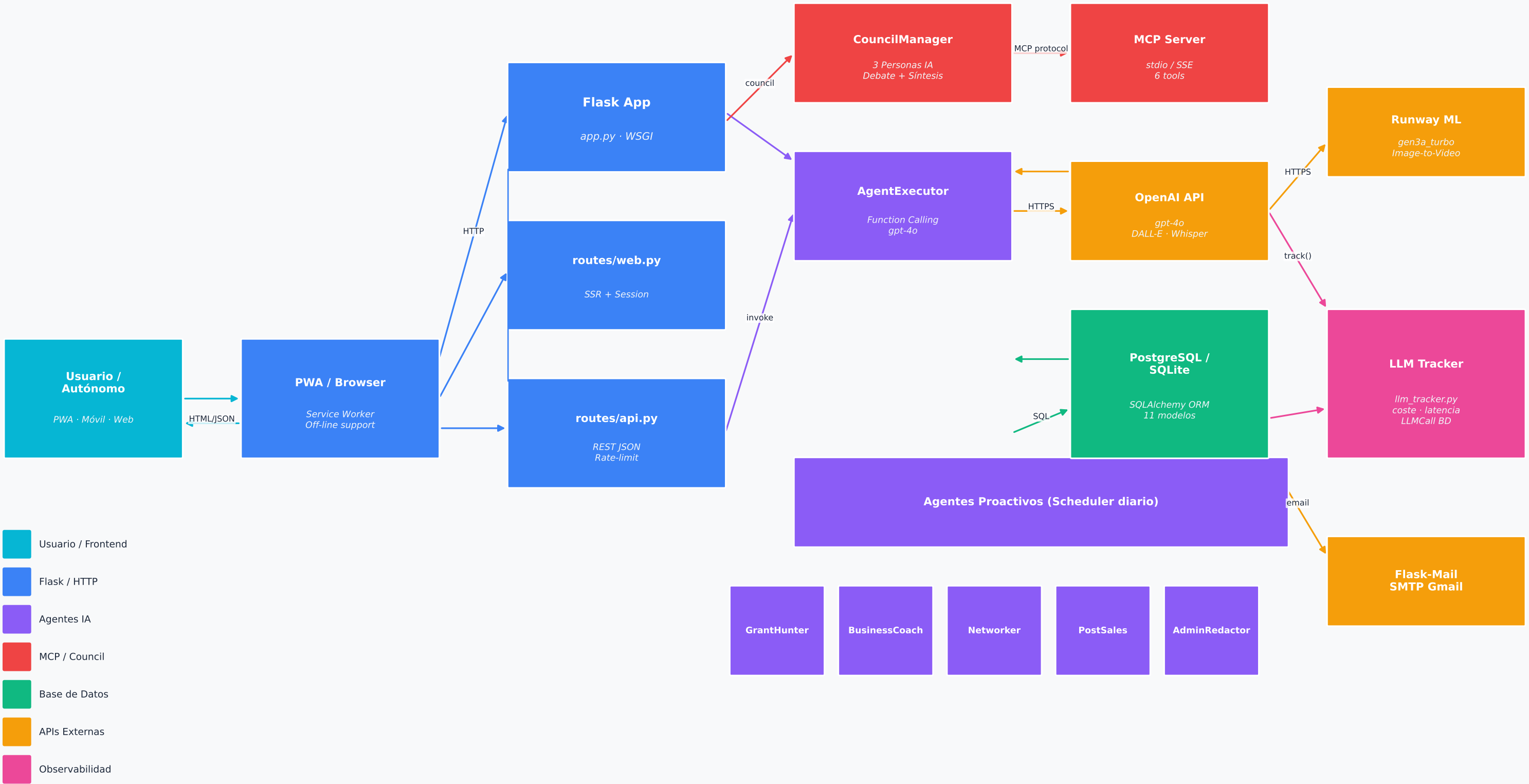
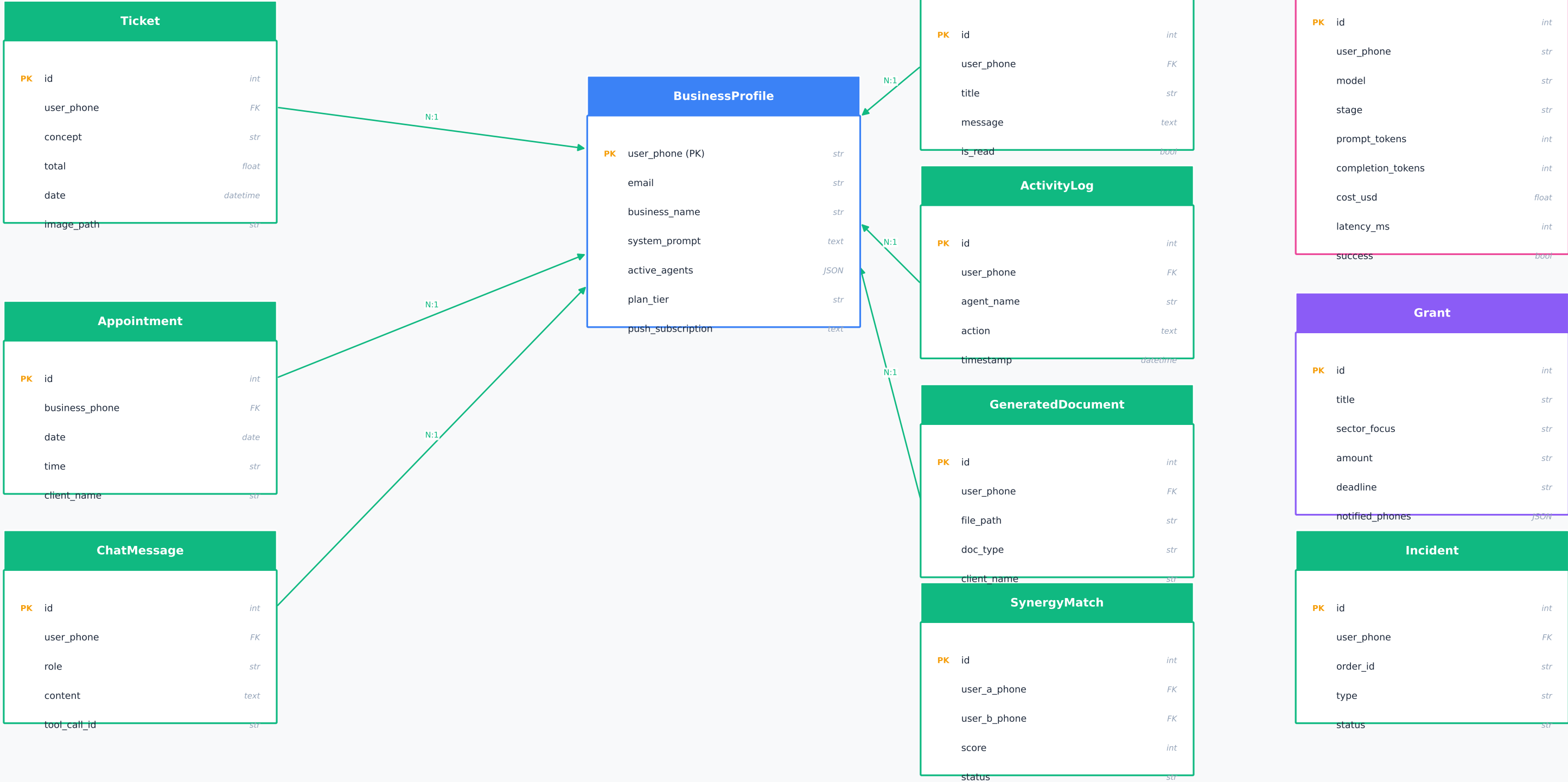


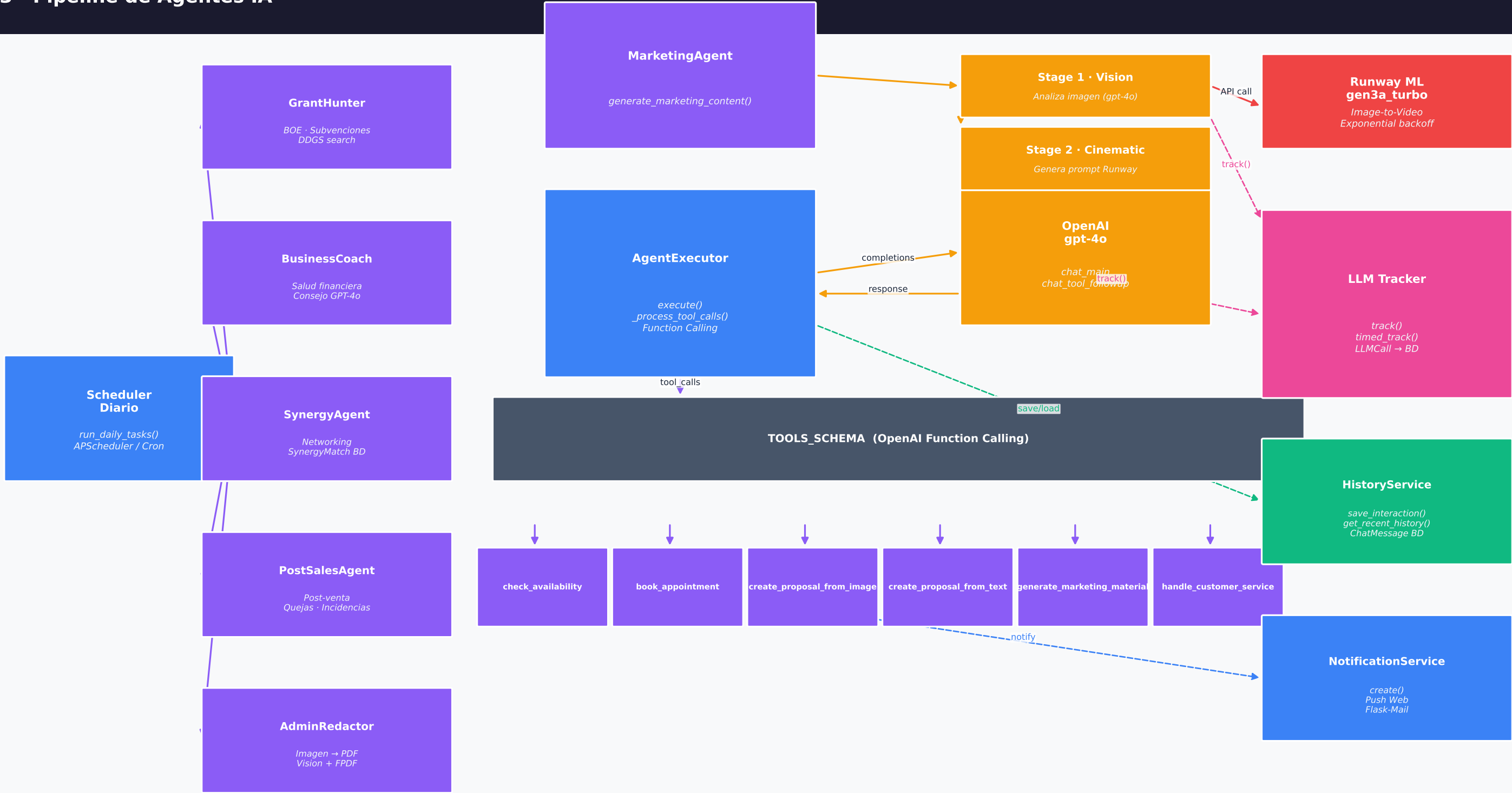
Ticketia / Zeptai

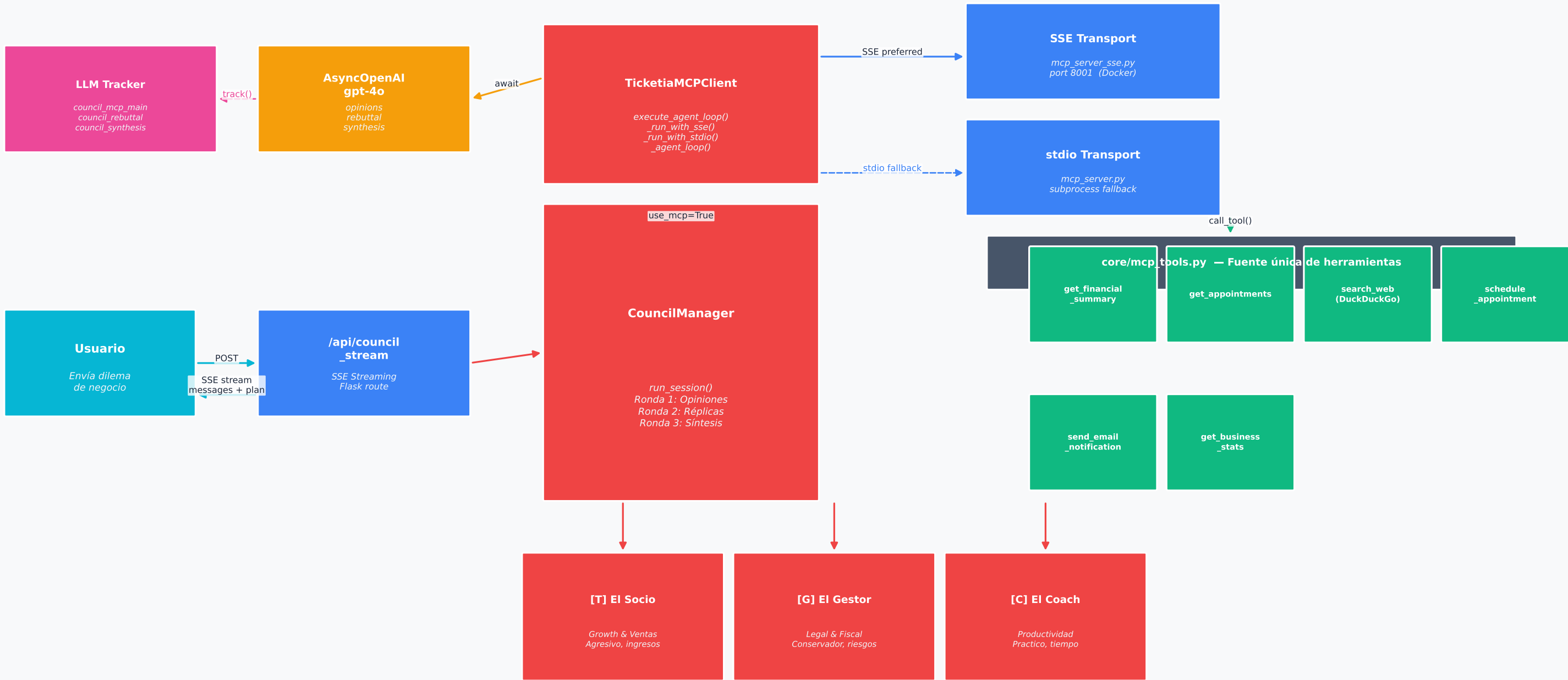
Arquitectura del Sistema

Asistente IA para Autónomos y PYMEs · Agentes Proactivos · MCP · LLM Observability









```
track(user_phone, model, stage,
      response, latency_ms)

timed_track(user_phone, model,
            stage)

_estimate_cost(model, in_tok,
               out_tok, extra)
```

llm_tracker.py

Modelo	Input / uso	Output
gpt-4o	\$2.50 / 1M in	\$10.00 / 1M out
gpt-4o-mini	\$0.15 / 1M in	\$0.60 / 1M out
dall-e-3	\$0.040 / imagen	—
whisper-1	\$0.006 / min	—
gen3a_turbo	\$0.05 / seg	—

LLMCall (db_models.py)

- user_phone · model · stage
- prompt_tokens · completion_tokens
- total_tokens · latency_ms
- cost_usd · success
- error_message · created_at

Stage	Módulo origen
chat_main	AgentExecutor
chat_tool_followup	AgentExecutor
video_analyze_image	MarketingAgent
video_generate_prompt	MarketingAgent
runway_video_generation	MarketingAgent
image_classify_intent	AdminRedactor
image_extract_data	AdminRedactor
council_mcp_main	MCPClient
council_mcp_followup	MCPClient
council_opinion	CouncilManager
council_rebuttal	CouncilManager
council_synthesis	CouncilManager

/metrics Dashboard

- Coste total estimado (USD) *SUM(cost_usd)*
- Tokens totales *SUM(total_tokens)*
- Latencia media *AVG(latency_ms)*
- Tasa de éxito *SUM(success)/COUNT(*)*
- Llamadas por modelo *GROUP BY model*
- Coste por modelo *GROUP BY model*
- Llamadas por stage *GROUP BY stage*
- Histórico 14 días *DATE(created_at)*

Chart.js CDN

*Line · Doughnut
Bar charts*

GET /api/metrics/llm

*JSON endpoint
Auth required*

Flask-Admin /admin

*Vista tabular LLMCall
Filtros · Exportar*



Tiempos típicos:

Auth + BD load: < 5 ms

GPT-4o (chat_main): 600 – 1500 ms

GPT-4o (tool followup): 400 – 900 ms

Runway video gen: 20 – 90 s (background)

AdminRedactor PDF: 1 – 3 s

Total (sin vídeo): 1 – 3 s